Teacher Wu Kuan | Midterm Must-Miss #1 — Nested Radicals: Big Minus Small

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1. The trap question

$\sqrt{(4 - 2\sqrt{3})} = ?$

2. The method

- Find two numbers with sum 4 and product 3 → 3 and 1.
- $4 - 2\sqrt{3} = (\sqrt{3} - 1)^2$, since $(\sqrt{3})^2 + 1^2 - 2\sqrt{3} = 4 - 2\sqrt{3}$.
- $\sqrt{(A^2)} = |A| \rightarrow \sqrt{3} - 1$ (big minus small), never $1 - \sqrt{3}$ (negative).

3. Parent check method

Ask your child to estimate the answer with a calculator; a negative result is an instant red flag.

Practice

1. $\sqrt{(5 - 2\sqrt{6})}$
2. $\sqrt{(7 + 2\sqrt{10})}$
3. $\sqrt{(6 - 2\sqrt{5})}$
4. $\sqrt{(8 - 2\sqrt{15})}$
5. $\sqrt{(4 + 2\sqrt{3})}$

Answers

1. $\sqrt{3} - \sqrt{2}$
2. $\sqrt{5} + \sqrt{2}$
3. $\sqrt{5} - 1$
4. $\sqrt{5} - \sqrt{3}$
5. $\sqrt{3} + 1$

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